

TP4- VOIP

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make menuselect

```

*****
Asterisk Module and Build Option Selection
*****

Press 'h' for help.

---> Add-ons (See README-addons.txt)
Applications
Bridging Modules
Call Detail Recording
Channel Event Logging
Channel Drivers
Codec Translators
Format Interpreters
Dialplan Functions
PBX Modules
Resource Modules
Test Modules
Compiler Flags
Utilities
AGI Samples
Core Sound Packages
Music On Hold File Packages
Extras Sound Packages

```

make

```
[LD] res_tonedetect.o -> res_tonedetect.so
Building Documentation For: channels pbx apps codecs formats cdr cel bridges funcs tests main res addons
+----- Asterisk Build Complete -----+
+ Asterisk has successfully been built, and +
+ can be installed by running:           +
+                                       +
+               make install              +
+-----+-----+

```

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make install

```
+----- Asterisk Installation Complete -----+
+
+   YOU MUST READ THE SECURITY DOCUMENT
+
+ Asterisk has successfully been installed.
+ If you would like to install the sample
+ configuration files (overwriting any
+ existing config files), run:
+
+ For generic reference documentation:
+   make samples
+
+ For a sample basic PBX:
+   make basic-pbx
+
+----- or -----+
+
+ You can go ahead and install the asterisk
+ program documentation now or later run:
+
+   make progdocs
+
+ **Note** This requires that you have
+ doxygen installed on your local system
+-----+
```

make samples

```
Updating asterisk.conf
/usr/bin/install -c -d "/var/spool/asterisk/voicemail/default/1234/INBOX"
build_tools/make_sample_voicemail "/var/lib/asterisk" "/var/spool/asterisk"
Installing file phoneprov/000000000000.cfg
Installing file phoneprov/000000000000-directory.xml
Installing file phoneprov/000000000000-phone.cfg
Installing file phoneprov/polycom_line.xml
Installing file phoneprov/polycom.xml
Installing file phoneprov/snom-mac.xml
root@Debian:~/asterisk-18.26.4#
```

make config

/etc/init.d/asterisk start

```
Starting asterisk (via systemctl): asterisk.service.
```

Sur quel(s) port(s) le service asterisk attend-il des requêtes ?

36109, 4569, 2727, 5000, 35201, 2000

Quelle(s) commande(s) avez-vous tapée(s) ?

ss -tuplen

```
root@Debian:~/asterisk-18.26.4# ss -tuplen
Netid State Recv-Q Send-Q Local Address:Port Peer Address:Port Process
udp UNCONN 0 0 0.0.0.0:36109 0.0.0.0:* users: (("asterisk",pid=37794,fd=9)) ino:37539 sk:1 cgroup:/system.slice/asterisk.service <->
udp UNCONN 0 0 0.0.0.0:4569 0.0.0.0:* users: (("asterisk",pid=37794,fd=14)) ino:37542 sk:2 cgroup:/system.slice/asterisk.service <->
udp UNCONN 0 0 0.0.0.0:2727 0.0.0.0:* users: (("asterisk",pid=37794,fd=11)) ino:37541 sk:3 cgroup:/system.slice/asterisk.service <->
udp UNCONN 0 0 0.0.0.0:5000 0.0.0.0:* users: (("asterisk",pid=37794,fd=18)) ino:37546 sk:4 cgroup:/system.slice/asterisk.service <->
udp UNCONN 0 0 [::]:35201 [::]:* users: (("asterisk",pid=37794,fd=10)) ino:37540 sk:5 cgroup:/system.slice/asterisk.service v6only:1 <->
tcp LISTEN 0 2 0.0.0.0:2000 0.0.0.0:* users: (("asterisk",pid=37794,fd=15)) ino:37543 sk:6 cgroup:/system.slice/asterisk.service <->
tcp LISTEN 0 128 0.0.0.0:22 0.0.0.0:* users: (("sshd",pid=682,fd=6)) ino:6476 sk:7 cgroup:/system.slice/ssh.service <->
tcp LISTEN 0 128 [::]:22 [::]:* users: (("sshd",pid=682,fd=7)) ino:6478 sk:8 cgroup:/system.slice/ssh.service v6only:1 <->
```

Observation 2 :

Combien de fichier contient-il ?

Le dossier contient 120 fichier

Combien de répertoire contient-il ?

Il ne contient pas de dossier

Le(s)quel(s) ?

```
neo@Debian:~$ ls /etc/asterisk/
acl.conf               config_test.conf      pjsip_wizard.conf
adsi.conf              console.conf           prometheus.conf
aeap.conf              dbsep.conf             queuerules.conf
agents.conf            dnsmgr.conf            queues.conf
alarmreceiver.conf     dsp.conf               res_config_mysql.conf
alsa.conf              dundi.conf             res_config_ldap.conf
amd.conf               enum.conf               res_config_odbc.conf
app_mysql.conf          extconfig.conf          res_config_sqlite3.conf
app_skel.conf           extensions.ael           res_config_sqlite.conf
ari.conf               extensions.conf          res_corosync.conf
ast_debug_tools.conf    extensions.lua           res_curl.conf
asterisk.adsi           extensions_minivm.conf   res_fax.conf
asterisk.conf           features.conf            res_http_media_cache.conf
calendar.conf           festival.conf            res_ldap.conf
ccss.conf              followme.conf            res_odbc.conf
cdr_adaptive_odbc.conf  func_odbc.conf           resolver_unbound.conf
cdr_banstalkd.conf      geolocation.conf         res_parking.conf
cdr.conf               hep.conf                 res_pgsql.conf
cdr_custom.conf          http.conf                res_pktccops.conf
cdr_manager.conf         iax.conf                 res_snmp.conf
cdr_mysql.conf           iaxprov.conf             res_stun_monitor.conf
cdr_odbc.conf            indications.conf          rtp.conf
cdr_pgsql.conf           logger.conf              say.conf
cdr_sqlite3_custom.conf  manager.conf             sip.conf
cdr_syslog.conf          meetme.conf              sip_notify.conf
cdr_tds.conf             mgcp.conf                skinny.conf
cel_banstalkd.conf       minivm.conf              sla.conf
cel.conf                misdns.conf              smdi.conf
cel_custom.conf          modules.conf              sorcery.conf
cel_odbc.conf            motif.conf                ss7.timers
cel_pgsql.conf           musiconhold.conf          stasis.conf
cel_sqlite3_custom.conf  muted.conf               statsd.conf
cel_tds.conf             ooh323.conf              stir_shaken.conf
chan_dahdi.conf          osp.conf                 telcordia-l.adsi
chan_mobile.conf         oss.conf                 test_sorcery.conf
cli_aliases.conf         phone.conf               udptl.conf
cli.conf                phoneprov.conf            unistim.conf
cli_permissions.conf     pjproject.conf           users.conf
codecs.conf              pjsip.conf               voicemail.conf
confbridge.conf          pjsip_notify.conf        vpb.conf
                           xmpp.conf
```

Quelle(s) commande(s) avez-vous tapée(s) ?

apt install tree

tree

```
1 directory, 120 files
```

ls -a -l (malgré le « 1 directory », les seuls dossier dans asterisk sont les dossiers . et ..)

```
drwxr-xr-x  2 root root  4096 10 nov.  17:00 .
drwxr-xr-x 70 root root  4096 10 nov.  17:22 ..
```

et ls -l

Observation 3 :

Que contient le fichier sip.conf ?

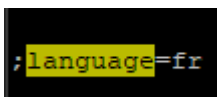
Il contient la configuration du protocole SIP

Que contient le fichier extensions.conf ?

Il contient la configuration des appels entrant et sortant

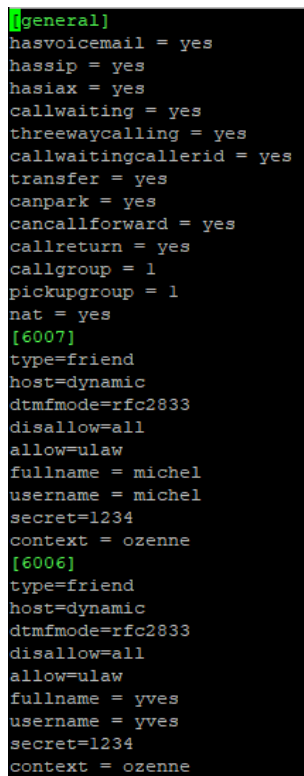
Observation 4 :

nano sip.conf



```
;language=fr
```

nano users.conf



```
[general]
hasvoicemail = yes
hassip = yes
hasiax = yes
callwaiting = yes
threewaycalling = yes
callwaitingcallerid = yes
transfer = yes
canpark = yes
cancallforward = yes
callreturn = yes
callgroup = 1
pickupgroup = 1
nat = yes
[6007]
type=friend
host=dynamic
dtmfmode=rfc2833
disallow=all
allow=ulaw
fullname = michel
username = michel
secret=1234
context = ozenne
[6006]
type=friend
host=dynamic
dtmfmode=rfc2833
disallow=all
allow=ulaw
fullname = yves
username = yves
secret=1234
context = ozenne
```

nano extensions.conf

```
[general]
static=yes
writeprotect=no
clearglobalvars=no

[globals]
SIP_6007=michel
SIP_6006=yves
CONSOLE=Console/dsp
IAXINFO=guest
TRUNK=DAHDI/G2
TRUNKMSD=1

[ozenne]
exten => 6006,1,Dial(PJSIP/michel,15)
exten => 6006,n,Hangup()

exten => 6007,1,Dial(PJSIP/yves,15)
exten => 6007,n,Hangup()
```

nano pjsip.conf

```
=====
; MICHEL
=====
[michel]
type=endpoint
aors=michel
auth=auth-michel
context=ozenne
disallow=all
allow=ulaw

[michel]
type=aor
max_contacts=1

[auth-michel]
type=auth
auth_type=userpass
username=michel
password=1234

=====
; YVES
=====
[auth-yves]
type=auth
auth_type=userpass
username=yves
password=1234

[yves]
type=aor
max_contacts=1

[yves]
type=endpoint
aors=yves
auth=auth-yves
context=ozenne
disallow=all
allow=ulaw
```

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Un des problèmes est que asterisk utilise extension.ael et pas le .conf. Donc les numéros de téléphones ne fonctionne pas.

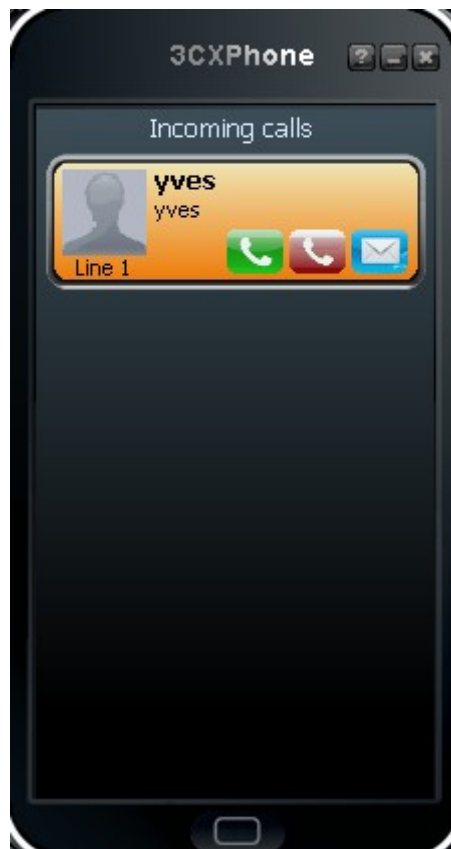
nano modules.conf

```
noload = app_voicemail_imap.so  
noload = app_voicemail_odbc.so  
noload => pbx_ael.so  
load => pbx_config.so
```

Dans le terminal de asterisk :

```
Debian*CLI> module unload pbx_ael.so
```

```
Debian*CLI> module load pbx_config.so
```



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Messagerie vocale :

extensions.conf

```
[ozenne]
exten => 6006,1,Dial(PJSIP/michel,15)

exten => 6007,1,Dial(PJSIP/yves,15)

exten => _6XXX,n,VoiceMail(${EXTEN}@ozenne)
exten => 600,1,VoiceMailMain(${CALLERID(num)}@ozenne)
```

voicemail.conf

```
[general]
[ozenne]
6007 => 1234,Michel
6006 => 1234,Yves
```



Avec un téléphone IP

Pour le téléphone IP, il faut le configurer sur l'interface graphique. Il faut mettre l'adresse IP du serveur sur Proxy et mettre les identifiants dans les informations juste en dessous.

Proxy:	172.16.155.68	
Outbound Proxy:		
Alternate Proxy:		
Alternate Outbound Proxy:		
Use Outbound Proxy:	no	Use OB Proxy In Dialog: yes
Register:	yes	Make Call Without Reg: no
Register Expires:	3600	Ans Call Without Reg: no
Use DNS SRV:	no	DNS SRV Auto Prefix: no
Proxy Fallback Intvl:	3600	Proxy Redundancy Method: Normal
Dual Registration:	no	
Display Name:	yves	User ID: yves
Password:	1234	Use Auth ID: yes
Auth ID:	yves	Reversed Auth Realm:
Mini Certificate:		
SRTP Private Key:		
Resident Online Number:		SIP URI:

Une fois appliqué, l'appel entre le téléphone IP et le softphone fonctionne.

